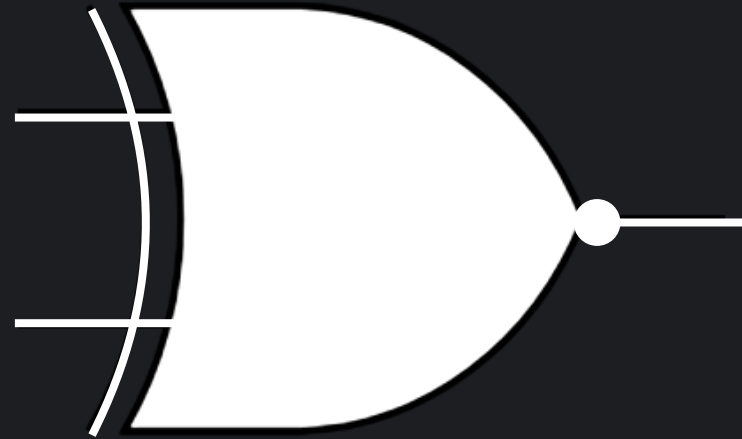


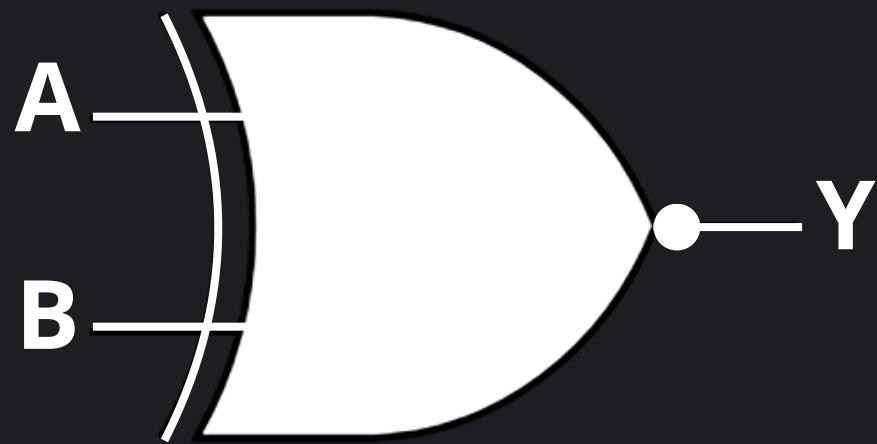
XNOR GATE



What is a XNOR Gate?

An XNOR gate is a basic logic gate that produces a HIGH (1) output when all inputs are the same (either all LOW or all HIGH). The output is LOW (0) when the inputs are different.

Truth Table



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

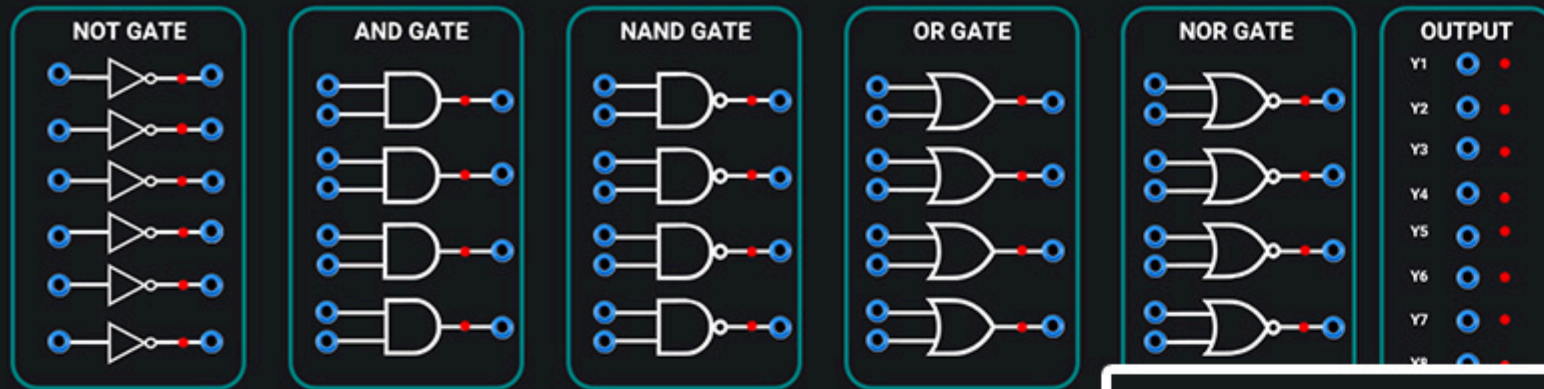
Function of the XNOR Gate

An XNOR gate performs a logical XNOR (Negated Exclusive OR) operation on its inputs. The main function of an XNOR gate is to combine two or more input signals and produce an output signal according to this rule:

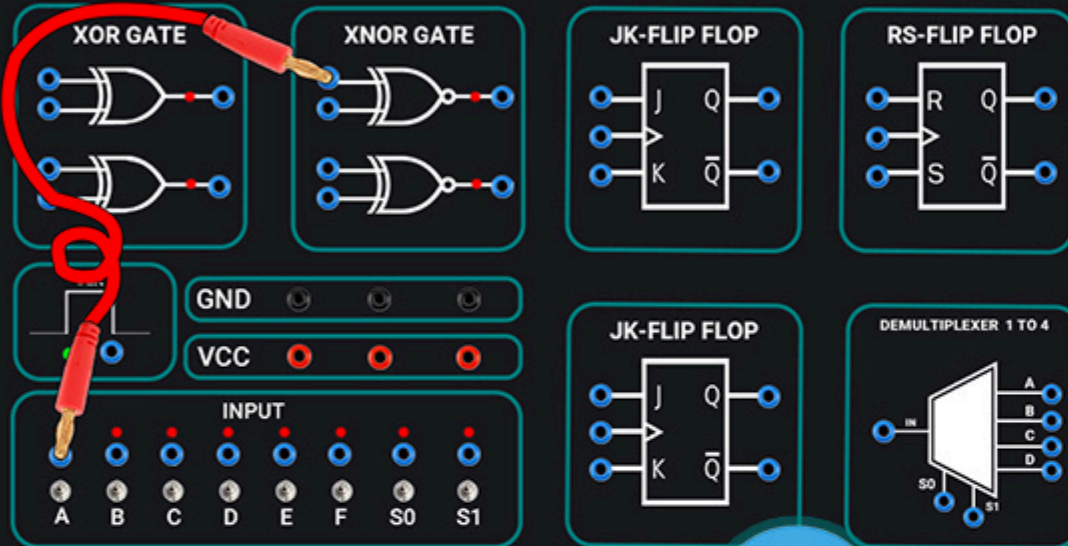
Output = 1 when all inputs are the same (either all 0 or all 1).

Output = 2 when the inputs are different.

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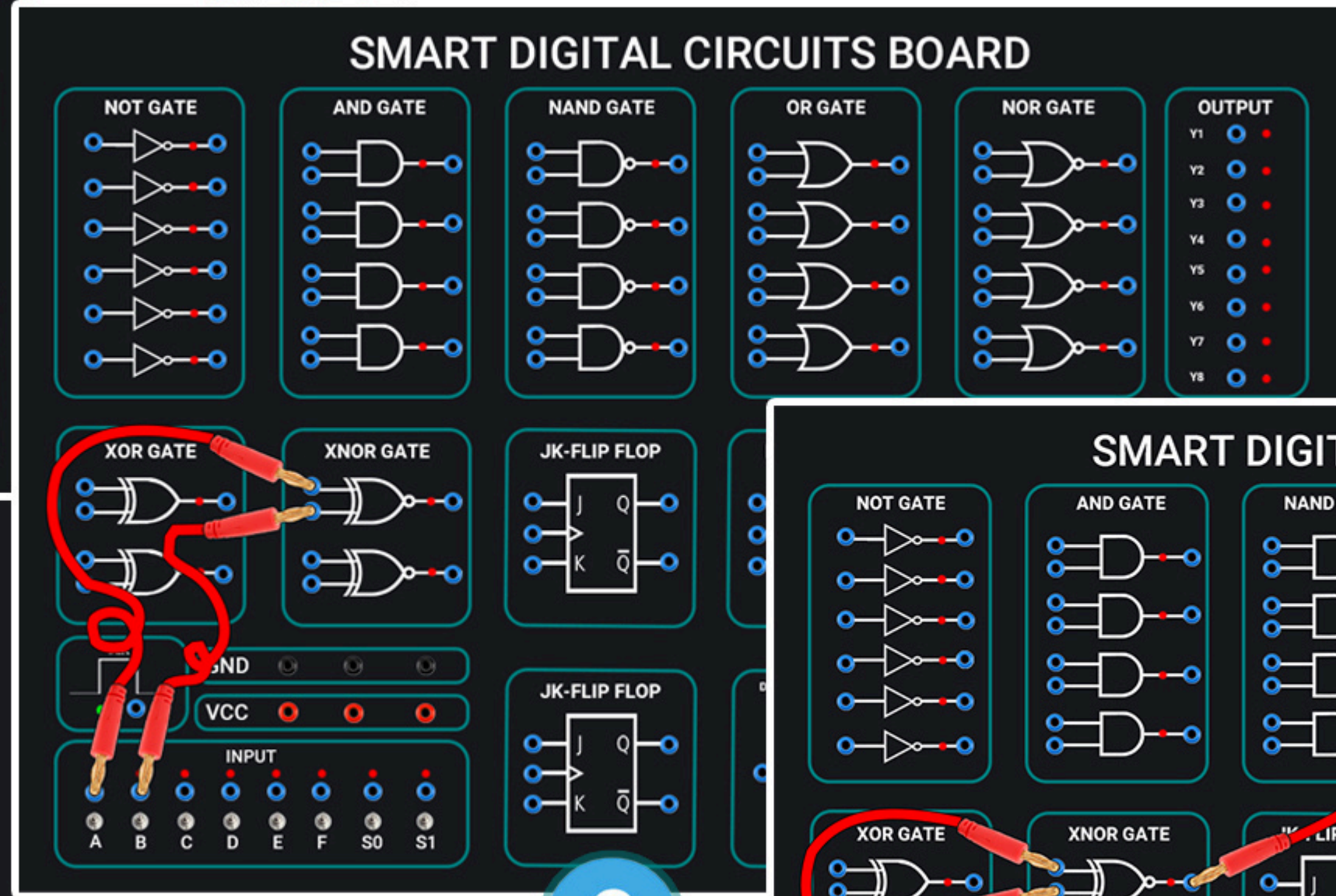


Connect the XNOR Gate



Connect a wire from the first input (top blue terminal) to switch A in the INPUT section.

1

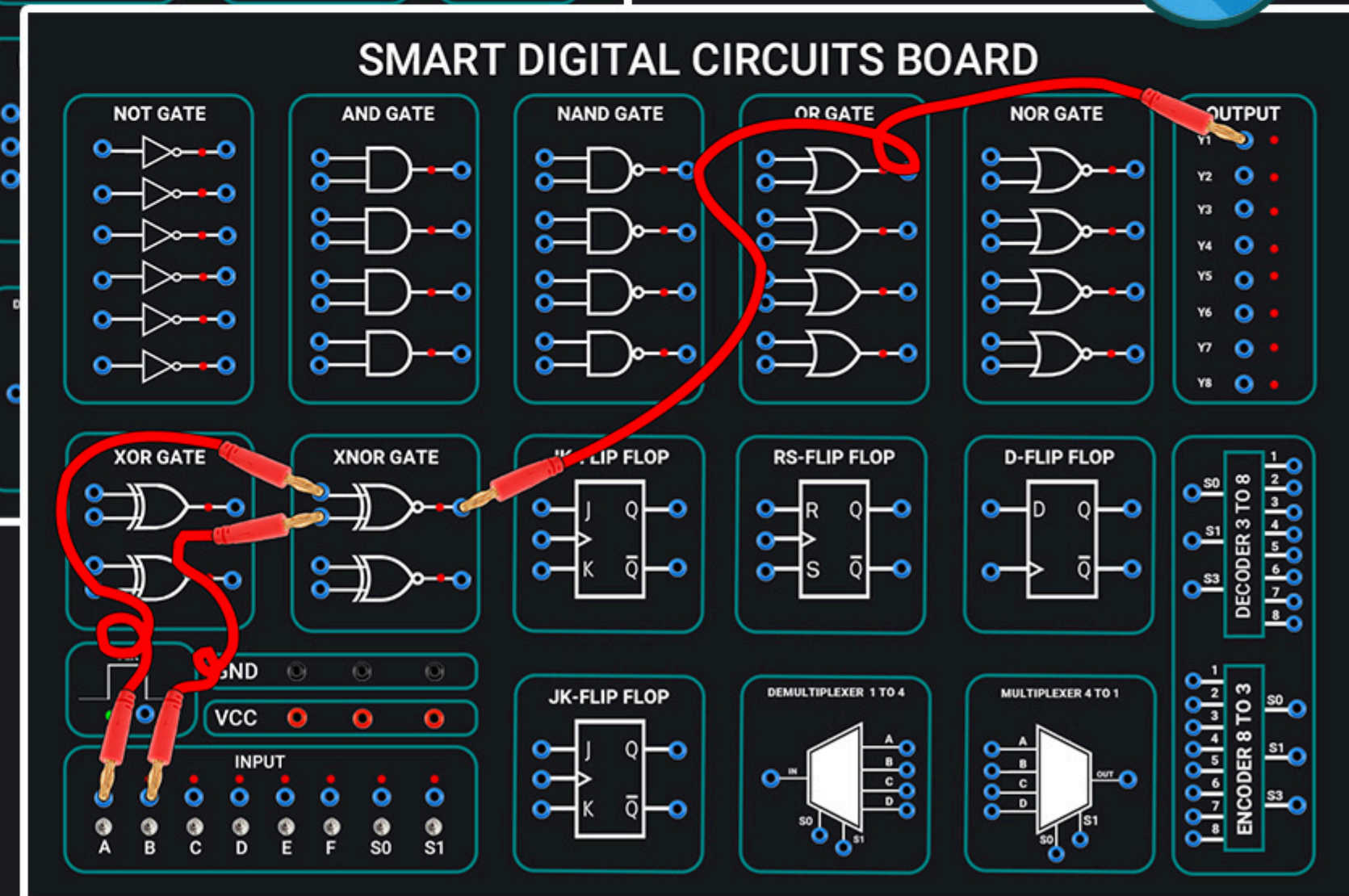


2

Connect a wire from the second input (bottom blue terminal) to switch B in the INPUT section

Connect a wire from the gate output to terminal Y1 (or any point from Y1 to Y8) in the OUTPUT section at the top right.

3



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